

Create University Database Design document

1. Department Table (departments)

Column Name	Data Type	Constraints on	Descripti
dept_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for each department
dept_name	VARCHAR(100)	NOT NULL, UNIQUE	Full name of the department
building	VARCHAR(50)	NOT NULL	Campus location or building name
budget	DECIMAL(12,2)	CHECK (budget > 0)	Annual department budget in ₹

2. Courses Table (courses)

Column Name	Data Type	Constraints	Descripti on
course_id	VARCHAR(10)	PRIMARY KEY	Unique academic code (e.g., CS101)
title	VARCHAR(100)	NOT NULL	Name of the course
credits	INT	CHECK (credits BETWEEN 1 AND 5)	Academic credit weight

	FOREIGN KEY	The
dept_id INT	references	offering
	departments (dept_id)	department

3. Student Table (students)

Column	Data Type	Constraints	Description Name
student_id	INT	PRIMARY KEY	Unique registration roll number
first_name	VARCHAR(50)	NOT NULL	First name of the student
last_name	VARCHAR(50)	NOT NULL	Last name/Surname of the student
email	VARCHAR(100)	NOT NULL, UNIQUE	Official university email address
phone	VARCHAR(15)	NOT NULL	Contact phone number
enroll_date	DATE	DEFAULT (CURRENT_DATE)	Date the student joined the university
dept_id	INT	FOREIGN KEY references departments(dept_id)	Major department

Students

table

data:

student_id	first_name	last_name	email	phone	enroll_date	dept_id
2026001	Sairam	Krishnan	sairam.k@univ.edu	9876543210	2026-06-01	101
2026002	Abami	Priya	abami.p@univ.edu	9876543211	2026-06-01	101
2026003	Rahul	Sharma	rahul.s@univ.edu	9876543212	2026-06-02	102
2026004	Meera	Nair	meera.n@univ.edu	9876543213	2026-06-03	103
2026005	Vikram	Singh	vikram.s@univ.edu	9876543214	2026-06-05	104

Department table data:

dept_id	dept_name	building	budget
101	Computer Science & Engineering	Ramanujan Block	1500000.00
102	Mechanical Engineering	Kalam Labs	1200000.00
103	Electrical & Electronics	Tesla Pavilion	950000.00
104	Business Administration	Aryabhata Hall	800000.00

Courses table data:

course_id	title	credits	dept_id
CS101	Introduction to Python	4	101
CS202	Database Management Systems 4		101

ME105	Engineering Graphics	3	102
EE210	Circuit Theory	4	103
BA101	Principles of Management	3	104